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$$f(x) = \ln \left(\tan \left(\frac{x}{2} + \frac{\pi}{4} \right) \right)$$

$$f'(x) = \frac{\sec^2 \left(\frac{x}{2} + \frac{\pi}{4} \right)}{2 \tan \left(\frac{x}{2} + \frac{\pi}{4} \right)}$$

$$f'(x) = \frac{1}{2 \cos \left(\frac{2x + \pi}{4} \right) \sin \left(\frac{2x + \pi}{4} \right)}$$

$$f'(x) = \frac{1}{\sin \left(x + \frac{\pi}{2} \right)}$$

$$f'(x) = \frac{1}{\cos(x)}$$

$$f'(x) = \sec(x)$$

References